

Grading — Specs Model

INFO 521 — Machine Learning Foundations

i Course-schedule placement

Reference document — applies to every homework unit, checkpoint, and project milestone; no single week.

Both projects use specifications grading. There are no points on milestones; each is judged against published criteria as **Satisfactory** or **Not-Yet**.

Milestones

- Every milestone lists its Satisfactory criteria at the top of the notebook. All criteria must be met — it is all-or-nothing, not partial credit.
- A **Not-Yet** returns with targeted feedback and **one revise cycle** back to Satisfactory. (Set a revise window that fits your cadence, e.g. one week.)
- A project is complete when all its milestones are Satisfactory **and** all its checkpoints are passed.

Checkpoints

- Short (~15–20 min), closed-book, in-class, one derivation each. Graded pass / retake — **retakeable** with an alternate version (provided in each checkpoint file).
- A milestone cannot be marked Satisfactory until its gating checkpoint is passed. This is what preserves the unaided-derivation signal the former exams carried, without a high-stakes timed event.
- Occupies the existing in-class-assessment slot.

Weighting

The two projects are **co-equal**. Project 2 is cumulative but is **not** weighted more heavily: specs grading already enforces that Project 2 mastery presupposes Project 1's foundations, so there is no need to also price that into a weight.

Revise-and-resubmit is the safety net; it replaces the former “final replaces lowest midterm” rule cleanly.

The exact contribution of each project to the final grade depends on how these fold into the broader course scheme (homework units, peer-engagement loops). Set that in the syllabus; this file governs the *mechanics*, not the percentages.

Bundling (suggested)

- **Project 1 bundle:** 1.1, 1.2, 1.3, 1.4 all Satisfactory + Checkpoints 1–3 passed.
- **Project 2 bundle:** two thematic gates + a capstone, all Satisfactory, plus Checkpoints 4 and 5 passed:

- **Gate A — Supervised:** 2.1 (approximate inference) + 2.2 (classification) Satisfactory, **and Checkpoint 4** (Newton–Raphson update) passed.
- **Gate B — Unsupervised:** 2.3 (clustering) + 2.4 (dimensionality reduction) Satisfactory, **and Checkpoint 5** (k-means objective / PCA) passed.
- **Capstone:** 2.5 (synthesis) Satisfactory, with its structured peer review completed.

Academic integrity note

Homework is open-book and AI-permitted in this course, so it certifies “can produce with help.” The checkpoints are the deliberate closed-book complement that certifies “can derive unaided.” Keep that division intact if you adjust the model.